Internal Transit: System Design Description

Document number	SHV-TR-SDD-0001
Revision	C
Status	APPROVED
Classification	INTERNAL
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Date	2024-08-16

1. System overview

Internal transit moves people and goods around the station: turbolift cars in vertical shafts and a tram that runs around every latitude band.

The system is led by Structural Engineering. It is delivered through 6 work packages with a combined budget of 52,848.18 Mcr.

2. Components

Code	Component	Specification	Drawing	Bill of materials
TR-TLC	Turbolift car	SHV-TR-SPEC-0101	SHV-TR-DWG-0201	SHV-TR-BOM-0301
TR-TLS	Turbolift shaft	SHV-TR-SPEC-0102	SHV-TR-DWG-0202	SHV-TR-BOM-0302
TR-TRM	Tram car	SHV-TR-SPEC-0103	SHV-TR-DWG-0203	SHV-TR-BOM-0303
TR-TRK	Tram track segment	SHV-TR-SPEC-0104	SHV-TR-DWG-0204	SHV-TR-BOM-0304

3. Architecture

3.1 Turbolift car

The turbolift car is specified in SHV-TR-SPEC-0101. Its governing parameter is the car capacity, set at 39 persons. The responsible engineer is Esme Hesk.

3.2 Turbolift shaft

The turbolift shaft is specified in SHV-TR-SPEC-0102. Its governing parameter is the shaft length, set at 8.66 km. The responsible engineer is Danna Tregar.

3.3 Tram car

The tram car is specified in SHV-TR-SPEC-0103. Its governing parameter is the seats, set at 86 ea. The responsible engineer is Elsk Redgrave.

3.4 Tram track segment

The tram track segment is specified in SHV-TR-SPEC-0104. Its governing parameter is the segment length, set at 68 m. The responsible engineer is Esme Hesk.

4. Interfaces

Document	Title
SHV-ICD-PD-TR	Interface Control Document: Power Distribution / Internal Transit
SHV-ICD-CN-TR	Interface Control Document: Computer Core & Data Network / Internal Transit

Document	Title
SHV-ICD-SF-TR	Interface Control Document: Superstructure & Frame / Internal Transit

5. Work packages

Work package	Name	Owner	Status	Complete
WP-SHV-TR-01	Internal Transit: Design & analysis	Kier Belcourt	IN_PROGRESS	76.5%
WP-SHV-TR-02	Internal Transit: Fabricate & install turbolift car	Vela Wade	IN_PROGRESS	62.9%
WP-SHV-TR-03	Internal Transit: Fabricate & install turbolift shaft	Tilda Delane	IN_PROGRESS	74.0%
WP-SHV-TR-04	Internal Transit: Fabricate & install tram car	Tilda Ashby	IN_PROGRESS	85.3%
WP-SHV-TR-05	Internal Transit: Fabricate & install tram track segment	Calla Arkwell	IN_PROGRESS	89.1%
WP-SHV-TR-06	Internal Transit: Integration & test	Calla Arkwell	IN_PROGRESS	77.2%

6. Risks

- Supply of turbolift car assemblies depends on a single supplier.
- Supply of turbolift shaft assemblies depends on a single supplier.
- Integration slots in the station schedule are shared with neighbouring zones.